

1. Team Ownership

The lab was completed as two independent implementations followed by a shared comparison. Naveen Aravapalli and Praveen Seera each built and evaluated their own models for all three tasks. We kept the code, checkpoints, outputs, metrics, and failure analyses separated by member in the repository, then compared the results as a team. This report focuses on what worked, where the models failed, and the trade-offs we observed between accuracy, efficiency, and reproducibility.

2. Task 1 - Language Modeling

2.1 Approach

Both members implemented character-level decoder-only GPT models from scratch in PyTorch. Neither implementation used `nn.Transformer`, `nn.MultiheadAttention`, or pretrained Transformer models. Both models used causal self-attention, learnable token and positional embeddings, feed-forward blocks, residual connections, LayerNorm, and autoregressive generation. The models intentionally differed in context length, width, depth, batch size, and seed.

Configuration	Naveen	Praveen
Vocabulary size	110	118
Context length	256	128
Embedding dimension	256	192
Attention heads	8	6
Dimension/head	32	32
Transformer blocks	4	3
FFN hidden dimension	1024	576
Dropout	0.10	0.15
Batch size	128	192
Initial learning rate	3e-4	4e-4
Warm-up	500 steps	400 steps
Optimizer	AdamW	AdamW
Scheduler	Warm-up + cosine	Warm-up + cosine
Epochs	10	10
Seed	171	2026
Parameters	3,278,336	1,183,104

2.2 Evaluation

Metric	Naveen	Praveen
Training cross-entropy	0.6190	0.7603
Validation cross-entropy	0.6285	0.7626
Generalization gap	0.0095	0.0022
Validation perplexity	1.8748	2.1438

Metric	Naveen	Praveen
Bits per character	0.9067	1.1001
Validation Top-1 accuracy	79.98%	75.89%
Distinct-1 (char)	0.0097	0.0093
Distinct-2 (char)	0.0824	0.0820
Distinct-3 (char)	0.2340	0.2463
Repeated 4-gram rate (char)	0.6072	0.5765
Average gradient norm	0.3488	0.4423
NaN count	0	0
Training throughput	909,419 tok/s	2,237,338 tok/s
Generation throughput	404.56 tok/s	702.83 tok/s
Peak GPU memory	3.57 GB	1.00 GB
Total training time	17.00 min	6.94 min

2.3 Comparative Analysis

The clearest Task 1 result was the trade-off between quality and efficiency. Naveen's larger GPT produced the better language-modeling scores: lower validation loss and perplexity, lower bits-per-character, and about four percentage points higher next-character accuracy. Praveen's model was much smaller and faster, using roughly one-third of the parameters and substantially less GPU memory. It also generated text faster and showed slightly less 4-gram repetition. In practice, the larger model bought better prediction quality, while the smaller model gave a stronger speed-and-memory profile.

2.4 Failure Analysis

Member	Representative failure	Interpretation
Naveen	"She saw a big tree with a long tail on it..."	Grammatically plausible but physically inconsistent; local fluency exceeded semantic grounding.
Naveen	"He knew that he would always pay for a day of step."	Well-formed syntax with an incoherent phrase.
Naveen	Repeated "She wanted to ... and see what was inside." templates	Greedy decoding collapsed toward high-probability TinyStories patterns.
Praveen	Repeated "Once upon a time ... Lily ..." passages	Strong greedy-decoding repetition / template collapse.
Praveen	"snacks on the wall"	Local grammaticality without coherent meaning.
Praveen	"It is a car full of trees. It is a toy can car."	Object/world-logic error.

Evidence: [task1_llm/naveen/results.md](#), [task1_llm/naveen/metrics_report.csv](#), [task1_llm/naveen/failure_analysis.md](#); [task1_llm/praveen/results.md](#), [task1_llm/praveen/metrics_report.csv](#), [task1_llm/praveen/failure_analysis.md](#).

3. Task 2 - Sentiment Classification

3.1 Shared Experimental Constraint

Each member trained one baseline and two experimental sentiment classifiers using embeddings learned from scratch, without pretrained embeddings or pretrained language models. Evaluation included discrimination, calibration, confidence intervals, paired significance testing, slice robustness, training efficiency, and a manual review of 20 errors.

3.2 Naveen Model Lineup

Model	Architecture	Emb.	Hidden	Dropout	Batch	LR	Params
Baseline GRU	1-layer uni-GRU	64	64	0.20	256	1e-3	3.91M
Experimental BiGRU	1-layer bidirectional GRU	128	128	0.30	256	5e-4	7.97M
Stacked BiGRU	2-layer BiGRU + MeanMax pooling	192	192	0.40	512	3e-4	12.92M

3.3 Praveen Model Lineup

Model	Architecture	Embedding	Sequence module / pooling	Params
Baseline CNN	2 Conv1D layers + max-over-time	128	Conv 128→128; kernels 5 and 3	8.55M
BiLSTM + Attention	Bidirectional LSTM + learned attention	160	hidden 128 each direction	10.82M
BiGRU + MaxPool	Bidirectional GRU + max pooling	128	hidden 96 each direction	8.54M

Praveen's final Task 2 documentation confirms max_len=256, batch size=256, AdamW with learning rate 2e-4, and seed 8511. The final executed notebook reuses the preserved checkpoints and outputs without retraining.

3.4 Core Test Performance

Member / Model	Accuracy	Macro-F1	ROC-AUC	PR-AUC	MCC	Brier ↓	ECE ↓
Naveen — GRU	0.94276	0.94276	0.98660	0.98696	0.88557	0.04345	0.01146
Naveen — BiGRU	0.94461	0.94460	0.98642	0.98614	0.88929	0.04274	0.01895
Naveen — 2L BiGRU	0.94613	0.94613	0.98814	0.98852	0.89229	0.04172	0.02112
Praveen — CNN	0.95068	0.95068	0.98914	0.98930	0.90149	0.03828	0.01311
Praveen — BiLSTM+Attn	0.95166	0.95166	0.99011	0.99044	0.90332	0.03683	0.01094
Praveen — BiGRU+Pool	0.94592	0.94592	0.98802	0.98840	0.89186	0.04124	0.01879

3.5 Efficiency

Member / Model	Training time	Train examples/s	Peak VRAM	Parameters
Naveen — GRU	2.83 min	9,302	0.14 GB	3.91M
Naveen — BiGRU	8.42 min	5,195	0.39 GB	7.97M
Naveen — 2L BiGRU	26.80 min	2,566	1.43 GB	12.92M
Praveen — CNN	8.68 min	7,743	0.21 GB	8.55M
Praveen — BiLSTM+Attn	23.75 min	2,829	1.05 GB	10.82M
Praveen — BiGRU+Pool	12.18 min	5,517	0.90 GB	8.54M

3.6 Statistical Testing and Robustness

Comparison	b	c	Exact p-value	Interpretation
Naveen GRU vs BiGRU	564	634	0.0462	Small but statistically significant improvement.
Naveen GRU vs 2L BiGRU	553	681	0.000297	Clear paired improvement over baseline.
Praveen CNN vs BiLSTM+Attn	767	804	0.3637	No significant paired difference.
Praveen CNN vs BiGRU+Pool	961	780	1.58e-05	Significant difference; BiGRU+Pool was worse overall.

3.7 Error Analysis Summary

Both 20-case reviews show that remaining errors are dominated by mixed/aspect-conflicting sentiment, late sentiment reversals, EDIT/UPDATE structures, negation, sarcasm, domain-specific wording, very short reviews, and long reviews with competing positive and negative evidence. Naveen's reviewed cases also exposed noisy labels and entity/target confusion; Praveen's cases similarly showed strong lexical bias, conditional sentiment, and long-review dilution.

Member	Example	Error type	Proposed direction
Naveen	"This place is so much better since they changed owners..." after a long negative history	Temporal sentiment shift	Recency-aware / sentence-level pooling
Naveen	"Food was good but, not great..." near 0.5	Balanced mixed sentiment	Calibration + sentence weighting
Praveen	"awesome" used conditionally/sarcastically	Sarcasm / conditional sentiment	Targeted sarcasm examples
Praveen	Very long hotel review with mixed billing/service details	Long-review truncation	Increase max_len / hierarchical pooling

3.8 Team-Level Task 2 Interpretation

The main Task 2 pattern was that extra complexity helped only when the architecture used it effectively. Praveen's BiLSTM with attention achieved the best overall result in the repository at 95.166% accuracy, with the CNN very close behind. Naveen's stacked BiGRU improved over his simpler GRU models, but the gain was modest compared with the extra training time. Praveen's BiGRU-MaxPool also finished below his simpler CNN. These results made the comparison useful: a larger recurrent model was not automatically better, and the strongest choices balanced sequence modeling with a simple, stable classification head.

Evidence: task2_sentiment/naveen/metrics_report.csv and outputs/; task2_sentiment/praveen/outputs/core_test_metrics.csv, bootstrap_95ci.csv, mcnemar_tests.csv, slice_robustness_metrics.csv, confusion matrices, manual_20_error_review.csv, and src/train_praveen.py.*

4. Task 3 - CycleGAN

4.1 Architecture and Training Setup

Configuration	Naveen	Praveen final
Generator	ResNet	ResNet
Residual blocks	9	9
Base channels	64	64
Upsampling	Transposed convolution	Nearest-neighbor + convolution

Configuration	Naveen	Praveen final
Discriminator	Single-scale 70×70 PatchGAN	Two-scale PatchGAN
Batch size	1	2
Adversarial loss	LSGAN / MSE	LSGAN / MSE
Cycle λ	10	10
Identity λ	5	5
Replay buffer	None	50 images
Learning rate	2e-4	2e-4
Adam betas	0.5, 0.999	0.5, 0.999
Final training point	Epoch 150	Epoch 300
LR schedule	Constant → linear decay	Constant → cosine decay
Seed	171	8511
GPU	RTX 4090	RTX 4090

Praveen's final selected configuration is taken from src/config_final.json and the final training/model code: 64 base channels, 9 residual blocks, a MultiScaleDiscriminator, no spectral normalization, LSGAN/MSE, Adam at 2e-4, cycle weight 10, identity weight 5, batch size 2, replay buffer 50, and 300 epochs. Earlier baseline settings are retained only as experiment history.

4.2 Distribution and Perceptual Metrics

Metric	Naveen	Praveen	Status note
FID A2B	109.4891	116.184	Praveen final
FID B2A	105.8679	109.365	Praveen final
Submission FID	107.6785	112.7744	Both final CSVs
MiFID A2B	0.43110	0.4335	Praveen final
MiFID B2A	0.41312	0.4129	Praveen final
Submission MiFID	0.42211	0.42320	Both final CSVs
KID A2B	0.02509	0.03749	Praveen value = local validation
KID B2A	0.01022	0.02039	Praveen value = local validation
Gen. precision A2B	0.6433	0.5333	Praveen value = local validation
Gen. recall A2B	0.3000	0.0519	Praveen value = local validation
Gen. precision B2A	0.3500	0.4882	Praveen value = local validation
Gen. recall B2A	0.6133	0.2033	Praveen value = local validation
Cycle L1 A	0.05189	0.12704	Praveen value = local validation
Cycle L1 B	0.06651	0.13638	Praveen value = local validation
LPIPS A	0.35315	0.59873	Praveen value = local validation
LPIPS B	0.30552	0.53714	Praveen value = local validation
Content cosine A2B	0.82844	0.82295	Praveen value = local validation

Metric	Naveen	Praveen	Status note
Content cosine B2A	0.78282	0.78939	Praveen value = local validation

4.3 Naveen Training Stability and Efficiency

Metric	Naveen	Praveen	Evidence / note
Parameter count	28,285,832	Not reported in recovered final-run metrics	Naveen: four networks
Training time	91.63 min	72.77 min	Praveen value is resumed final segment only, not full run
Throughput	16.81 images/s	Not recovered	Naveen epoch-150 logged throughput
Peak VRAM	2.45 GiB	Not recovered	Final-run Praveen value unavailable
Generator gradient norm	23.90	Not recovered	Naveen epoch-150
Discriminator gradient norm	9.79	Not recovered	Naveen epoch-150
NaN / Inf	0 / 0	Not recovered	No unsupported value inferred for Praveen

4.4 Kaggle Result

The recorded Kaggle team was PairProgramming_Team_25. The best displayed leaderboard result was rank 48 with score -54.0503. The leaderboard value is reported at team level and is not attributed to either member individually.

Item	Recorded value
Team	PairProgramming_Team_25
Leaderboard rank	48
Best displayed score	-54.0503
Reporting scope	Team-level Kaggle result
Naveen local submission metrics	FID 107.6785; MiFID 0.42211
Praveen local submission metrics	FID 112.7744; MiFID 0.42320

4.5 Analysis

The biggest Task 3 takeaway was that training longer and adding a more complex discriminator did not automatically improve image quality. Naveen's epoch-150 CycleGAN achieved the better final FID and combined submission result. His precision/recall values also showed a clear directional difference: Monet-to-Photo outputs were more precise but covered less of the target distribution, while Photo-to-Monet outputs covered more of the distribution at lower precision. Praveen's final model used a multi-scale discriminator, replay buffer, larger batch size, nearest-neighbor upsampling, and 300 epochs, but its final FID remained higher. This was a useful reminder that GAN quality depends on the whole training balance, not simply on more epochs or architectural complexity.

4.6 Human Audit

Both members completed a 30-sample, two-rater human audit. Naveen scored artifact quality with higher values meaning better quality, while Praveen scored artifact severity with lower values meaning fewer visible artifacts, so the artifact means are not directly comparable. The manual review generally supported the automated results: content was preserved well in many samples, while style consistency and visible artifacts remained the main weaknesses. This matched what we saw in the generated examples, especially in difficult textures and stronger style transfers.

Human-audit metric	Naveen	Praveen	Interpretation
Samples / raters	30 / 2	30 / 2	15 A2B + 15 B2A; two independent raters

Human-audit metric	Naveen	Praveen	Interpretation
Mean style score	3.667 / 5	4.017 / 5	Higher is better
Mean content score	4.033 / 5	4.250 / 5	Higher is better
Artifact score	3.433 / 5 quality	3.933 / 5 severity	Scales differ: Naveen higher=better; Praveen lower=better
Style weighted kappa	0.671	0.720	Inter-rater agreement
Content weighted kappa	0.742	0.368	Inter-rater agreement
Artifact weighted kappa	0.610	0.769	Inter-rater agreement
Mean exact agreement	57.78%	73.33%	Praveen mean of 76.67%, 63.33%, 80.00%

Evidence: [task3_gan/naveen/metrics_report.csv](#), [full_metrics_report.csv](#), [results.md](#), [training/reproducibility manifests](#), [submission.csv](#); [task3_gan/praveen/results.md](#), [full_metrics_report.csv](#), [submission.csv](#), [src/config_final.json](#), [src/train.py](#), [src/models.py](#).

5. Reproducibility

5.1 Repository Structure and Evidence Trail

GitHub: <https://github.com/aravapallinaveen/DATA266-Lab1-Pair25>

The GitHub repository keeps each member's work under separate task folders and includes source code, metrics, outputs, failure analyses, and reproducibility records. Large model checkpoints are included in the Canvas submission package when they are too large for GitHub. The final Task 3 evidence notebooks were saved with visible outputs: Naveen's final metrics notebook has all 16 code cells executed, and Praveen's final evidence notebook has all 7 code cells executed.

Evidence category	Representative repository path
Task 1 Naveen	task1_llm/naveen/
Task 1 Praveen	task1_llm/praveen/
Task 2 Naveen	task2_sentiment/naveen/
Task 2 Praveen	task2_sentiment/praveen/
Task 3 Naveen	task3_gan/naveen/
Task 3 Praveen	task3_gan/praveen/
Raw logs / manifests	reproducibility/raw_logs/ and reproducibility/manifests/
Final report	report/DATA266_Lab1_Report_Team_25.pdf

5.2 Hardware Disclosure

Task / Member	Recorded hardware
Task 1 — Naveen	NVIDIA GeForce RTX 5090
Task 1 — Praveen	NVIDIA GeForce RTX 5090
Task 2 — Naveen	AMD Ryzen 9 7950X + NVIDIA GeForce RTX 4090 (23.99 GB)
Task 2 — Praveen	Intel Core i7-10750H @ 2.60 GHz + NVIDIA GeForce RTX 2060
Task 3 — Naveen	NVIDIA GeForce RTX 4090; CUDA 13.0
Task 3 — Praveen	NVIDIA GeForce RTX 4090 (RunPod final run)

5.3 Checkpoint Traceability

The Canvas submission ZIP contains the executed notebooks and saved model weights required for each part. The report links back to the GitHub repository for code and evidence, while the large Task 3 checkpoints are carried in the submission package.

Naveen Task 3 final checkpoint SHA-256: [9cc42d24ca52fd53d3ba1b690c92a5fb0fd4eb8a27dfd087f86e030a3ae24152](#)

Before export, the report was checked against the latest repository artifacts. Naveen's Task 3 checkpoint is checksum-traceable. Praveen's final 300-epoch checkpoint is included with the submission package. The original console log for Praveen's final 300-epoch segment was not recovered, so we did not reconstruct or invent it; the available earlier raw logs were preserved unchanged and the limitation is stated openly.

6. Overall Findings

Across the three tasks, the strongest model was not always the largest or longest-trained model. In Task 1, Naveen's larger GPT gave better predictive quality, while Praveen's smaller GPT was substantially more efficient. In Task 2, Praveen's BiLSTM-attention model achieved the best classification accuracy, while Naveen's experiments showed a clear capacity-versus-cost progression. In Task 3, Naveen's CycleGAN achieved the better final FID/MiFID, even though Praveen used a longer 300-epoch schedule and a more complex discriminator.

The most useful lesson was that evaluation needs more than a single score. We used efficiency measurements, statistical tests, failure analysis, human review, and reproducibility evidence to understand why models behaved differently. The remaining weaknesses were also consistent across the tasks: limited semantic/world knowledge in generated text, mixed or compositional sentiment in reviews, and style-versus-content trade-offs in image translation.

7. References

Vaswani, A., et al. (2017). Attention Is All You Need. Advances in Neural Information Processing Systems.

Eldan, R., & Li, Y. (2023). TinyStories: How Small Can Language Models Be and Still Speak Coherent English?

Zhu, J.-Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. ICCV.

Appendix A - Task 1 Full Metrics

Metric	Naveen	Praveen
Train CE	0.6190347	0.7603223
Validation CE	0.6284905	0.7625633
Perplexity	1.8747784	2.1437643
Bits/character	0.9067201	1.1001463
Generalization gap	0.0094558	0.0022410
Validation Top-1	0.7998204	0.7588916
Distinct-1 char	0.0096872	0.0092836
Distinct-2 char	0.0823577	0.0819540
Distinct-3 char	0.2339996	0.2463154
Repeated 4-gram char	0.6072294	0.5765347
Distinct-1 word	0.2448980	0.2837573
Distinct-2 word	0.6848249	0.7414300
Distinct-3 word	0.8685492	0.9068627
Repeated 4-gram word	0.0711501	0.0304220
Avg gradient norm	0.3487685	0.4422888
NaN count	0	0
Parameters	3,278,336	1,183,104
Training tokens/s	909,419	2,237,338
Generation tokens/s	404.56	702.83

Metric	Naveen	Praveen
Peak GPU memory	3.569 GB	1.005 GB
Training time	16.9996 min	6.9389 min

Appendix B - Task 2 Full Metrics

B.1 Precision / Recall / F1 (all averaging modes)

Model	P-macro	R-macro	F1-macro	P-micro	R-micro	F1-micro	P-weighted	R-weighted	F1-weighted
N-GRU	0.94281	0.94276	0.94276	0.94276	0.94276	0.94276	0.94281	0.94276	0.94276
N-BiGRU	0.94468	0.94461	0.94460	0.94461	0.94461	0.94461	0.94468	0.94461	0.94460
N-2L	0.94615	0.94613	0.94613	0.94613	0.94613	0.94613	0.94615	0.94613	0.94613
P-CNN	0.95080	0.95068	0.95068	0.95068	0.95068	0.95068	0.95080	0.95068	0.95068
P-BiLSTM	0.95166	0.95166	0.95166	0.95166	0.95166	0.95166	0.95166	0.95166	0.95166
P-BiGRU	0.94594	0.94592	0.94592	0.94592	0.94592	0.94592	0.94594	0.94592	0.94592

B.2 Confusion Matrices

Model	TN	FP	FN	TP
N-GRU	17,818	1,182	993	18,007
N-BiGRU	17,822	1,178	927	18,073
N-2L BiGRU	18,045	955	1,092	17,908
P-CNN	17,909	1,091	783	18,217
P-BiLSTM+Attn	18,119	881	956	18,044
P-BiGRU+Pool	17,915	1,085	970	18,030

B.3 95% Bootstrap Confidence Intervals

Model	Accuracy 95% CI	Macro-F1 95% CI	MCC 95% CI
N-GRU	[0.94029, 0.94524]	[0.94029, 0.94523]	[0.88061, 0.89052]
N-BiGRU	[0.94242, 0.94713]	[0.94242, 0.94713]	[0.88494, 0.89434]
N-2L	[0.94397, 0.94850]	[0.94396, 0.94850]	[0.88796, 0.89701]
P-CNN	[0.94839, 0.95294]	[0.94839, 0.95293]	[0.89691, 0.90599]
P-BiLSTM	[0.94979, 0.95370]	[0.94979, 0.95370]	[0.89959, 0.90740]
P-BiGRU	[0.94381, 0.94811]	[0.94381, 0.94811]	[0.88763, 0.89624]

B.4 Slice Robustness

Naveen — Macro-F1 / Error Rate

Slice	GRU	BiGRU	2L BiGRU
Short ≤ 64	0.94461 / 0.05502	0.94668 / 0.05293	0.94734 / 0.05237
Medium 65–128	0.93748 / 0.06189	0.94029 / 0.05913	0.94239 / 0.05692
Long ≥ 129	0.93451 / 0.06098	0.93155 / 0.06314	0.93858 / 0.05664
Has negation	0.93853 / 0.05956	0.94148 / 0.05673	0.94318 / 0.05487
No negation	0.93156 / 0.05079	0.93017 / 0.05168	0.93188 / 0.05108

Praveen — Macro-F1 / Error Rate

Slice	CNN	BiLSTM+Attn	BiGRU+Pool
Short ≤ 64	0.95310 / 0.04592	0.95177 / 0.04733	0.94565 / 0.05326
Medium 65–128	0.95166 / 0.04829	0.95387 / 0.04604	0.94751 / 0.05241

Slice	CNN	BiLSTM+Attn	BiGRU+Pool
Long ≥ 129	0.94034 / 0.05715	0.94421 / 0.05310	0.93953 / 0.05769

Appendix C - Task 3 Detailed Metrics and Audit

Naveen's final automated metric set includes FID/MiFID, KID, generative precision/recall, cycle L1, LPIPS, content cosine, stability measurements, parameter count, training time, throughput, and peak VRAM. Praveen's final 300-epoch FID/MiFID are reported from the final evaluator; KID, precision/recall, LPIPS, cycle L1, and content-cosine values in his metrics file are explicitly retained as local-validation measurements. Both members completed the required 30-sample, two-rater human audit.